

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - V

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Class :SY IT	Batch : 1
Date of Assignment : 12/08/26	Date/Time of Submission :12/08/26

PART – B Practical Questions:

5a. Write a python code to demonstrate importing pandas libraries and create data frame objects.

Code:

```
import pandas as pd

data = {
    "name": ["Shiv", "Shivay", "Shivam"],
    "marks": [50, 40, 30],
    "qualification": ["Pass", "Fail", "Pass"]
}
df = pd.DataFrame(data)

newdf = df.filter(items=["name", "marks", ])
print(newdf)
print("\nAmit_S057")
```

Output:

```
===== RESTART: C
=
   name marks
0  Shiv   50
1 Shivay  40
2 Shivam  30
```

OR

Code:

```
import pandas as pd

data = {
    "name": ["Shiv", "Shivay", "Shivam"],
    "marks": [50, 40, 30],
    "qualification": ["Pass", "Fail", "Pass"]
}
df = pd.DataFrame(data)

newdf = df.filter(items=["name", "marks", "qualification"])
print(newdf)
print("\nAmit_S057")
```

Output:

```
===== RESTART: C:/Users/
=
   name marks qualification
0  Shiv   50         Pass
1 Shivay   40         Fail
2 Shivam   30         Pass
```

5b. Write a python code to show statistical information on a given data set.

Code:

i. Statistics.mean

```
# Import statistics Library
import statistics as st

# Calculate average values
print(st.mean([1, 3, 5, 7, 9, 11, 13]))
print(st.mean([1, 3, 5, 7, 9, 11]))
print(st.mean([-11, 5.5, -3.4, 7.1, -9, 22]))
```

Output:

```
===== RESTART: C
7
6
1.8666666666666667
```

ii.Statistics.median

Code:

```
# Import statistics Library
import statistics as st

# Calculate middle values
print(st.median([1, 3, 5, 7, 9, 11, 13]))
print(st.median([1, 3, 5, 7, 9, 11]))
print(st.median([-11, 5.5, -3.4, 7.1, -9, 22]))
print("\nAmit_S057")
|
```

Output:

```
===== RESTART: C:
7
6.0
1.05

Amit_S057
|
```

iii. Statistics.mode

Code:

```
# Import statistics Library
import statistics as st

# Calculate the mode
print(st.mode([1, 3, 3, 3, 5, 7, 9, 11]))
print(st.mode([1, 1, 3, -5, 7, -9, 11]))
print(st.mode(['red', 'green', 'blue', 'red']))
print("\nAmit_S057")
|
```

Output:

```
===== RESTART: C:/Users/mvlui
==
3
1
red

Amit_S057
|
```

iv. Statistics.stdev

Code:

```
# Import statistics Library
import statistics as st

# Calculate the standard deviation from a sample of data
print(st.stdev([1, 3, 5, 7, 9, 11]))
print(st.stdev([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print(st.stdev([-11, 5.5, -3.4, 7.1]))
print(st.stdev([1, 30, 50, 100]))
print("\nAmit_S057")
|
```

Output:

```
===== RESTART: C:/Users/m
==
3.7416573867739413
0.6925797186365383
8.414471660973927
41.67633221226007

Amit_S057
|
```

v. Statistics.variance

Code:

```
# Import statistics Library
import statistics

# Calculate the variance from a sample of data
print(statistics.variance([1, 3, 5, 7, 9, 11]))
print(statistics.variance([2, 2.5, 1.25, 3.1, 1.75, 2.8]))
print(statistics.variance([-11, 5.5, -3.4, 7.1]))
print(statistics.variance([1, 30, 50, 100]))
print("\nAmit_S057")
```

Output:

```
===== RESTART: C:/User
==
14
0.47966666666666663
70.80333333333333
1736.9166666666667

Amit_S057
```

5c. Write a python code to create a pandas series from dictionaries.

1.

Code:

```
# import the pandas lib as pd
import pandas as pd

# create a dictionary
dictionary = {'D': 70, 'B': 60, 'C': 50}

# create a series
series = pd.Series(dictionary)

print(series)
print("\nAmit_S057")
```

Output:

```
===== RESTART:
=====
D    70
B    60
C    50
dtype: int64
Amit_S057
|
```

2.

Code:

```
# import the pandas lib as pd
import pandas as pd

# create a dictionary
dictionary = {'A': 750, 'B': 100, 'C': 90}

# create a series
series = pd.Series(dictionary, index=['C', 'A', 'B'])

print(series)
print("\nAmit_S057")
```

Output:

```
===== RESTART: C:/Us
==
C    90
A   750
B   100
dtype: int64

Amit_S057
|
```

5d. Write a python code to demonstrate filter pandas series with Boolean arrays.

```
# importing pandas as pd
import pandas as pd

# dictionary of lists
dict = {'name':["amit", "payal", "shivam", "prachi"],
        'degree': ["BSC IT", "BCOM", "M.Tech", "MBA"],
        'score':[90, 40, 80, 98]}

df = pd.DataFrame(dict, index = [True, False, True, False])
print(df)
print("\nAmit_S057")
```

Output:

```
===== RESTART: C:/Users/i
==
      name degree score
True  amit  BSC IT   90
False payal  BCOM   40
True  shivam M.Tech   80
False prachi  MBA   98

Amit_S057
|
```

PART – B Practical Questions:

- 5a. Write a python code to demonstrate importing pandas libraries and create data frame object.
- 5b. Write a python code to show statistical information on given data set.
- 5c. Write a python code to create pandas series from dictionaries.
- 5d. Write a python code to demonstrate filter pandas series with Boolean arrays.

PART – C Curiosity Question: